
COCONUT on Small Reasoning Models and Its Potential

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Abstract

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1 Introduction

The Chain of Continuous Thought (COCONUT) [Hao et al., 2024] explores the potential of the latent layer of reasoning, as an alternate to traditional CoT [Wei et al., 2022]. Its prior work has shown that the latent layer of continuous reasoning can reduce unnecessary statements, and focus on the core of the problem.

In January, the DeepSeek team proposed a distillation-based method for implementing a reasoning model via reinforcement learning [DeepSeek-AI, 2025], and they applied the approach to the Qwen-2.5B [Yang et al., 2024] to a reasoning model. Given the limitations of a 2.5B parameter model in reasoning tasks, we aim to assess COCONUTS potential in enhancing its performance.

Fine-Tuning We are using SFT to adapt the model for COCONUT, which involves `<sot>` and `<eot>` special tokens as the start and end of the reasoning process, instead of the original `<think>` and `</think>`. (The DeepSeek-R1’s reproduction indicates that it’s just a tag instead of special token.) Since the reasoning procedure is totally hidden, we should use a defined token to determine whether the chain should be terminated or not, instead of wrapping to show it to user.

Actual Training Then, we are leveraging the demonstrated model to fine-tune them to COCONUT progressively, increasing latent reasoning layer at most as possible, through supervised data and reinforcement learning. We prepared Open Thoughts datasets [Team, 2025] to provide the fine-tuning data.

Evaluation We employ a difficulty-ascending ranking to evaluate its capability, particularly on mathematics. We prepared math problems from SAT, AMC, AIME, to Gaokao mathematics, to Olympics, etc. to find the maximum potential.

To improve evaluation accuracy, we also introduced the `<soa>` (Start of Answer) and `<eoas>` (End of Answer) tokens to indicate the answer, replacing the commonly used format—using `\boxed` to wrap the answer. Then we can use simple string matching to evaluate the model.

References

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